

Evaluation of Feature Selection Using Wrapper For Numeric Dataset With Random Forest Algorithm

Arie Nugroho

Faculty of Computer Science
Dian Nuswantoro University
Semarang, Indonesia
arie2020udinus@gmail.com

Ahmad Zainul Fanani

Faculty of Computer Science
Dian Nuswantoro University
Semarang, Indonesia
a.zainul.fanani@dsn.dinus.ac.id

Guruh Fajar Shidik

Faculty of Computer Science
Dian Nuswantoro University
Semarang, Indonesia
guruh.fajar@research.dinus.ac.id

Abstract—Preprocessing is more than half of machine learning process. Dimensionality reduction is one of the preprocessing task, which included feature extraction and selection. Feature selection used for identify relevant and remove not relevant feature. The goal of this research is to select relevant feature using wrapper method for early diabetes prediction dataset which has been transformed to numeric dataset previously. Forward and backward selection are used in wrapper method, that's combine with random forest and cross validation. Random forest is decision tree enhancement, which is group of trees that can produce difference or same result at each tree. The most results are made as final result. The final result from feature selection with wrapper method can make higher accuracy than without feature selection for numeric dataset and the number of feature can be reduced. With features selection which is sequential forward selection it has 98.84 % accuracy with 11 feature selected and with sequential backward selection, it has 99.03 % accuracy with same number of features selected. With reduced features, will reduces complexity of trees and time required in mining process.

Keywords—wrapper, feature selection, forward selection, backward selection, random forest, diabetes

I. INTRODUCTION

Human activities produce a large variety and size of data, which will be useless if not processed properly. With machine learning existing dataset that has been produced in the past can be processed so that it becomes useful information and knowledge for humans to support their activities in the future. The problem is it's not always in a form that is ready to be processed directly, so that pre-processing data is required. Data preprocessing has a very important role in machine learning. The dataset needs to be preprocessed before being processed with a machine learning algorithm. Machine learning method has supervised and unsupervised learning [1]. In supervised learning, the dataset used which has a label to help the prediction [1]. In classification task, dataset should has a label besides features. Medical data sets are one example of many cases that can be used in machine learning. In this study, using early diabetes prediction datasets. With machine learning and existing datasets, it can be predicted that a person has diabetes or not. One of pre-processing tasks is reduction [2]. Dimensionality reduction is reducing dimension or feature from dataset[3]. Dimensionality reduction consists of feature extraction and selection[2]. Feature extraction is produce new

feature which is extracted from original features in dataset[4]. Feature selection is process to identify relevant feature over not relevant, redundant feature form a dataset[4]. Feature selection has three sub task are filter, wrapper and embedded method[5]. Wrapper identify feature with predictive accuracy by cross validation. The model was trained use chosen feature incrementally. It need more time, because inspect one by one training and testing with cross validation[4]. Random forest is merging or combination of several decision trees by conducting training on the processed data. To make a model, training dataset split to several part to build every decision tree. After models is made, then use data testing to test each decision tree. Then each decision tree provides a vote based on their models. The highest number of votes was used as a prediction[6]. In this research discuss about usage of forward selection and backward elimination as wrapper method. The purpose of this study is to select relevant feature with forward and backward elimination for random forest algorithm with diabetes prediction dataset.

II. RELATED WORK

Benbelkacem and Atmani [6] propose comparison random forest with other machine learning algorithms for pima Indians dataset. The result is Random Forest has smallest error rate over other algorithms with 0.21 at 40 trees. Xu and Wang[7] propose to construct model to predict accuracy of diabetes mellitus diagnosis with Random Forest, XG-Boost algorithm and weighted feature selection which has result 93.75 %. Gupta and Goel [8] develop prediction model with KNN for best performance. It combine with feature selection and normalization also find optimal number of neighbors with 85.06 % accuracy. Kumar [9] propose random forest for early prediction of diabetes dataset which the accuracy is greater than other machine learning algorithms . Dai, Chen, Zhu and Zhang [10] discuss about the ensemble machine learning of random forest which was produced classification result accurately which is using breast cancer dataset and has 98 % accuracy. Those related works, using all features from dataset. In this research using feature selection is used to select which features are related to the class and to improve accuracy.

III. METHODOLOGY

This research propose wrapper method to identify selected features which are used from dataset and combine it with

random forest to calculate best accuracy. Features selection is very important to find feature which necessary from dataset. This because not all features will have significant influence for model. But doesn't mean it will be perfect for all a class, it depends to entirely data that used. This research first step is read or load original dataset, then transform all features from nominal to integer or numeric data type. After that, then split dataset to features and label. Next step is features selection with wrapper method. This step is to select features which will produce higher accuracy. Feature selection in this research with wrapper to combine forward or backward selection with random forest to get necessary feature to increase accuracy. Research steps for this study shown at figure 1.

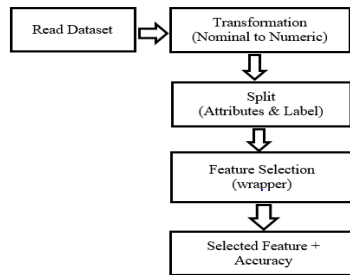


Fig. 1. Proposed workflow

A. Dataset Description

This study uses early stage diabetes risk prediction dataset which is one of the public dataset from University of California Irvine (UCI) repository. It has 16 features, 1 label or class and 520 instances or record, with no missing values. Features description of original dataset shown at table 1.

TABLE I. FEATURE DESCRIPTION

Features		Data Type
Number	Name	
0	Age	Numeric
1	Sex	Nominal (male or female)
2	Polyuria	Nominal (yes or no)
3	Polydipsia	Nominal (yes or no)
4	Sudden weight loss	Nominal (yes or no)
5	Weakness	Nominal (yes or no)
6	Polyphagia	Nominal (yes or no)
7	Genital Thrush	Nominal (yes or no)
8	Visual Blurring	Nominal (yes or no)
9	Itching	Nominal (yes or no)
10	Irritability	Nominal (yes or no)
11	Delayed Healing	Nominal (yes or no)
12	Partial Paresis	Nominal (yes or no)
13	Muscle	Nominal

Features		Data Type
	Stiffness	(yes or no)
14	Alopecia	Nominal (yes or no)
15	Obesity	Nominal (yes or no)
16	Class	Nominal (positive or negative)

Age feature has numeric data type, which mean age of patient. The other features is nominal data type which has yes or no value. Those features are explain signatures of diabetes disease. The label is nominal data type with positive or negative values, which is positive means that the patient has diabetes disease prediction while negative means has not.

B. Transformation and Split

The transformation process is used to transform all feature values form nominal to numeric data type. This step used for normalize the features, cause random forest can't process nominal data type. After transformation process, next process is splitting dataset to features and label. The features are number 1 until 15 and label is number 16. The features name shown at table 1.

C. Random Forest

It was defined as a group of decision trees or forest which is model was built randomly from feature[11]. This algorithm very suitable for dataset that has numeric type. The more feature used, more variety of trees there are. Steps for this algorithm shown at figure 2.

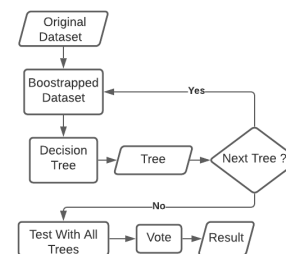


Fig. 2. Random Forest Steps

First steps from random forest is create a bootstrapped dataset randomly. Second is create a decision tree using the bootstrapped dataset but only use a random subset of feature at each step. Third, back to first then second step and keep repeat it until numbers of tree. Using a bootstrapped sample and considering only a subset of the feature at each step results in a wide variety of trees. The variety is what makes random forest more effective than individual decision trees. For testing, take a case or sample then run it down to the every tree that made. Each tree will produce a result or which call a vote. After running the case down all of the trees in random forest, the most vote will concluded as the final result or conclusion for that case.

D. Feature Selection

The feature selection method used in this research is the wrapper, which is greed search techniques, they are sequential forward selection (SFS) and sequential backward selection (SBS) [12]. It's running and testing by 10 fold cross validation, so every value at each feature will be used as data training and testing. It use 10 fold because based on experiments which are repeated 8 times, start with 3 until 10 fold and has same result [13]. The second reasons is 10 fold used to optimize random forest [14]. With wrapper, the feature selection is executed while looking at the result of the running learning algorithm [15] which is random forest. Feature selection with wrapper method and random forest process shown at figure 3.

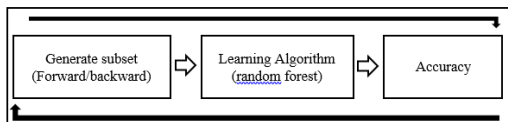


Fig. 3. Wrapper with random forest

Wrapper feature selection method steps are generate subset or features from normalized dataset with forward (SFS) or backward (SBS) while monitoring the accuracy using cross validation with learning algorithm which is random forest, it just looping until all features are processed, that's explain how wrapper can make high accuracy but it needs more time than filter method. Features selected one by one until all features are processed and then calculate the accuracy. With SFS algorithm, the step is select feature one by one forward while calculate accuracy with chosen algorithm until all features from dataset selected. Accuracy's value will increase or decrease in every feature selected. With SBS algorithm, the steps are select all features first then eliminate feature one by one backward while seeing accuracy with random forest.

IV. EXPERIMENT AND RESULT

This section explain about experiment and result comparison between without feature selection and with feature selection method. Feature selection using wrapper method which are SFS and SBS.

A. Read and transformation dataset

This section explain about original dataset shown at table 2 and transformation result from nominal to numeric data type shown at table 3

TABLE II. ORIGINAL DATASET

Age	Gender	Polyuria		class
40	Male	No		Positive
58	Male	No		Positive
....				
42	Male	No	...	Negative

The original dataset has 17 features based on table 1. First feature, which is age has numeric data type and others features has binominal data type or nominal with 2 values only.

TABLE III. TRANSFORMED DATASET

Age	Gender	Polyuria		class
16	0	0		1
34	0	0		1
....				
18	0	0		0

The transformed dataset was processed from original dataset, which is age feature also transformed to smaller value in numeric data type. Gender feature the values was male then transform to 0 (zero). Class feature which has positive value transformed to 1 and when has negative value transformed to 0.

B. Split dataset

This step is to split transformed dataset to features and label. The features begin from age until obesity and the class as label. Based on table 1, feature number 0 to 15 and feature number 16 as label.

C. Feature Selection

Random forest algorithm with 10 folds cross validation and without feature selection has 97,5 % mean cross validation or accuracy which is it takes all feature from dataset. With feature selection using wrapper method and random forest as learning algorithm. SFS produce 98,84 % accuracy with 11 features selected which is shown at figure 3. SBS produce 99,03 % accuracy with 11 features which is shown at figure 4.

	feature_idx	cv_scores	avg_score	feature_names
1	(2,)	[0.7884615384615384, 0.8076923076923077, 0.75, ...]	0.823077	(polyuria,)
2	(1, 2)	[0.7884615384615384, 0.9807692307692307, 0.961, ...]	0.871154	(gender, polyuria)
3	(1, 2, 3)	[0.8461538461538461, 0.9615384615384615, 0.961, ...]	0.894231	(gender, polyuria, polydipsia)
4	(1, 2, 3, 11)	[0.8461538461538461, 0.9423076923076923, 1.0, ...]	0.917308	(gender, polyuria, polydipsia, dh)
5	(1, 2, 3, 11, 14)	[0.8461538461538461, 0.9230769230769231, 0.989, ...]	0.942308	(gender, polyuria, polydipsia, dh, alopecia)
6	(0, 1, 2, 3, 11, 14)	[0.9615384615384615, 0.9230769230769231, 0.989, ...]	0.953846	(ac, gender, polyuria, polydipsia, dh, alopecia)
7	(0, 1, 2, 3, 7, 11, 14)	[0.9615384615384615, 0.9615384615384615, 0.980, ...]	0.973077	(ac, gender, polyuria, polydipsia, gt, dh, alo)
8	(0, 1, 2, 3, 7, 8, 11, 14)	[0.9615384615384615, 0.9615384615384615, 0.980, ...]	0.976923	(ac, gender, polyuria, polydipsia, gt, vb, dh, ...)
9	(0, 1, 2, 3, 7, 8, 11, 13, 14)	[0.9615384615384615, 0.9615384615384615, 0.980, ...]	0.980769	(ac, gender, polyuria, polydipsia, gt, vb, dh, ...)
10	(0, 1, 2, 3, 7, 8, 9, 11, 13, 14)	[0.9615384615384615, 0.9615384615384615, 1.0, ...]	0.986538	(ac, gender, polyuria, polydipsia, gt, vb, dh, ...)
11	(0, 1, 2, 3, 7, 8, 9, 11, 13, 14, 15)	[0.9807692307692307, 0.9807692307692307, 1.0, ...]	0.988462	(ac, gender, polyuria, polydipsia, gt, vb, dh, ...)
12	(0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 13, 14, 15)	[0.9615384615384615, 0.9615384615384615, 1.0, ...]	0.988462	(ac, gender, polyuria, polydipsia, gt, vb, dh, ...)
13	(0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 13, 14, 15)	[0.9615384615384615, 0.9807692307692307, 0.980, ...]	0.984615	(ac, gender, polyuria, polydipsia, svl, gt, vb, ...)
14	(0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15)	[0.9615384615384615, 0.9807692307692307, 0.961, ...]	0.982692	(ac, gender, polyuria, polydipsia, svl, gt, vb, ...)
15	(0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14, ...)	[0.9615384615384615, 0.9807692307692307, 0.961, ...]	0.982692	(ac, gender, polyuria, polydipsia, svl, polyph, ...)

Fig. 4. SFS process

SFS process beginning with one feature first which is polyuria, with learning algorithm and 10 fold cross validation, which is produce 82 % accuracy. After polyuria, second feature is processed is gender and polyuria, there are produce 87 %. The next process is to proceed to other features one by one and keep count the accuracy. Highest accuracy seen when feature number is 11 which has 98 % and not increasing with the rest features. That's mean that the rest of features which are feature number 12 until 15 do not really have a significant relationship with the class. With eleven features from SFS process can make highest accuracy and reduce the number of features, which will automatically reduce the time required for learning and testing process.

	feature_idx	cv_scores	avg_score	feature_names
16	(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, ...)	[0.9815384615384615, 0.9807692307692307, 0.980...	0.980769	(ac, gender, polyuria, polydipsia, swl, weakne...
15	(0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ...)	[0.9815384615384615, 0.9807692307692307, 1.0, ...	0.988482	(ac, gender, polyuria, polydipsia, weakness, p...
14	(0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15)	[0.9815384615384615, 0.9807692307692307, 1.0, ...	0.988482	(ac, gender, polyuria, polydipsia, weakness, p...
13	(0, 1, 2, 3, 6, 7, 8, 9, 10, 12, 13, 14, 15)	[0.9815384615384615, 0.9807692307692307, 0.980...	0.988482	(ac, gender, polyuria, polydipsia, polyphagia...
12	(0, 1, 2, 3, 7, 8, 9, 10, 12, 13, 14, 15)	[0.9815384615384615, 1.0, 0.9807692307692307, 1.0, ...	0.988482	(ac, gender, polyuria, polydipsia, gt, vb, itc...
11	(0, 1, 2, 3, 7, 8, 9, 10, 13, 14, 15)	[0.9815384615384615, 1.0, 1.0, 1.0, 1.0, ...	0.990385	(ac, gender, polyuria, polydipsia, gt, vb, itc...
10	(0, 1, 2, 3, 7, 8, 9, 10, 14, 15)	[0.9807692307692307, 0.9807692307692307, 1.0, ...	0.982692	(ac, gender, polyuria, polydipsia, gt, vb, itc...
9	(0, 1, 2, 3, 7, 8, 9, 10, 14)	[0.9807692307692307, 0.9807692307692307, 1.0, ...	0.982692	(ac, gender, polyuria, polydipsia, gt, vb, itc...
8	(0, 1, 2, 3, 7, 9, 10, 14)	[0.9807692307692307, 0.9807692307692307, 1.0, ...	0.984615	(ac, gender, polyuria, polydipsia, gt, itching, ir...
7	(0, 1, 2, 3, 9, 10, 14)	[0.9807692307692307, 0.9807692307692307, 0.981...	0.971154	(ac, gender, polyuria, polydipsia, itching, ir...
6	(0, 1, 2, 3, 10, 14)	[0.9807692307692307, 0.9807692307692307, 0.981...	0.983482	(ac, gender, polyuria, polydipsia, irritabilit...
5	(0, 1, 2, 10, 14)	[0.9807692307692307, 0.9807692307692307, 0.981...	0.951923	(ac, gender, polyuria, irritability, alopecia)
4	(0, 1, 2, 10)	[0.9807692307692307, 0.9807692307692307, 0.981...	0.930769	(ac, gender, polyuria, irritability)
3	(0, 1, 2)	[0.9807692307692307, 0.9807692307692307, 0.942...	0.884615	(ac, gender, polyuria)
2	(1, 2)	[0.9807692307692307, 0.9807692307692307, 0.981...	0.871154	(gender, polyuria)
1	(2)	[0.9807692307692307, 0.9807692307692307, 0.75...	0.823077	(polyuria)

Fig. 5. SBS process

SBS process beginning with all features with learning algorithm and 10 fold cross validation, which is produce 98 % accuracy. Next step is reducing one by one feature and keep count the accuracy. Start with all features it has 98 % accuracy. Next step is with 15 features processed and it has higher accuracy. Highest accuracy seen when feature number is eleven, which has 99 %. That's mean that features from all until twelve do not really have a significant relationship with the class. With eleven features from SBS beside SFS process, it's can make highest accuracy and reduce the number of features, so automatically reducing time required for training and testing.

D. Selected Feature and Accuracy

This step explain about comparison result between using feature selection with wrapper method and without feature selection (FS) for early diabetes prediction dataset using random forest algorithm shows that there is an increase in accuracy using feature selection rather than not. Comparison result shown at figure 6.

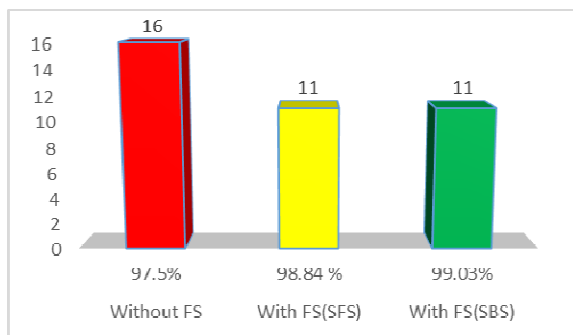


Fig. 6. Comparison Result

Without feature selection has 97.5 % accuracy with all features which is depicted with red bar. With feature selection, which is has eleven features only can make higher accuracy which has 98.84 % accuracy, which is depicted as yellow and

green bar which has 99.03 % accuracy. SFS with random forest has higher accuracy than just using random forest. Features selection using SBS and random forest has higher accuracy than SFS which is need same number of features.

V. CONCLUSION

Feature selection is important for make better accuracy for early stage diabetes risk prediction dataset and random forest classifier algorithm, without feature selection, it has 97.50 % accuracy with all feature selected. Using feature selection which is sequential forward selection it has 98.84 % accuracy with 11 Feature selected. Using sequential backward selection, it has 99.03 % accuracy with 11 features selected. Further work is to comparing wrapper method with another method, such as filter or embedded and use two or more dataset for comparison.

ACKNOWLEDGMENT

Credit to University of California Irvine (UCI) data repository which has provided public dataset and Dian Nuswantoro University.

REFERENCES

- [1] R. Saravanan and P. Sujatha, "Algorithms: A Perspective of Supervised Learning Approaches in Data Classification," *2018 Second Int. Conf. Intell. Comput. Control Syst.*, no. Iciccs, pp. 945–949, 2018.
- [2] R. Vaishali, R. Sasikala, S. Ramasubbareddy, S. Remya, and S. Nalluri, "Genetic algorithm based feature selection and MOE Fuzzy classification algorithm on Pima Indians Diabetes dataset," *Proc. IEEE Int. Conf. Comput. Netw. Informatics, ICCNI 2017*, vol. 2017-Janua, pp. 1–5, 2017, doi: 10.1109/ICCNI.2017.8123815.
- [3] M. Vikram, R. Pavan, N. D. Dineshbhai, and B. Mohan, "Performance evaluation of dimensionality reduction techniques on high dimensional data," *Proc. Int. Conf. Trends Electron. Informatics, ICOEI 2019*, vol. 2019-April, no. Icoei, pp. 1169–1174, 2019, doi: 10.1109/icoei.2019.8862526.
- [4] N. Gopika and A. E. A. Meena Kowshalaya, "Correlation Based Feature Selection Algorithm for Machine Learning," *Proc. 3rd Int. Conf. Commun. Electron. Syst. ICCES 2018*, no. Icces, pp. 692–695, 2018, doi: 10.1109/CESYS.2018.8723980.
- [5] S. Egea, A. Rego Manez, B. Carro, A. Sanchez-Esguevillas, and J. Lloret, "Intelligent IoT traffic classification using novel search strategy for fast-based-correlation feature selection in industrial environments," *IEEE Internet Things J.*, vol. 5, no. 3, pp. 1616–1624, 2018, doi: 10.1109/IIOT.2017.2787959.
- [6] S. Benbelkacem and B. Atmani, "Random forests for diabetes diagnosis," *2019 Int. Conf. Comput. Inf. Sci. ICCIS 2019*, pp. 1–4, 2019, doi: 10.1109/ICCISci.2019.8716405.
- [7] Z. Xu and Z. Wang, "A Risk prediction model for type 2 diabetes based on weighted feature selection of random forest and xgboost ensemble classifier," *11th Int. Conf. Adv. Comput. Intell. ICACI 2019*, pp. 278–283, 2019, doi: 10.1109/ICACI.2019.8778622.
- [8] S. C. Gupta and N. Goel, "Enhancement of Performance of K-Nearest Neighbors Classifiers for the Prediction of Diabetes Using Feature Selection Method," *2020 IEEE 5th Int. Conf. Comput. Commun. Autom. ICCCA 2020*, pp. 681–686, 2020, doi:

10.1109/ICCCA49541.2020.9250887.

- [9] K. Vijiyakumar, B. Lavanya, I. Nirmala, and S. Sofia Caroline, "Random forest algorithm for the prediction of diabetes," *2019 IEEE Int. Conf. Syst. Comput. Autom. Networking, ICSCAN 2019*, pp. 1–5, 2019, doi: 10.1109/ICSCAN.2019.8878802.
- [10] B. Dai, R. C. Chen, S. Z. Zhu, and W. W. Zhang, "Using random forest algorithm for breast cancer diagnosis," *Proc. - 2018 Int. Symp. Comput. Consum. Control. IS3C 2018*, pp. 449–452, 2019, doi: 10.1109/IS3C.2018.00119.
- [11] C. C. Aggarwal, *Data Mining The Textbook*, vol. 14, no. 3. 1981.
- [12] N. K. Suchetha, A. Nikhil, and P. Hrudya, "Comparing the wrapper feature selection evaluators on twitter sentiment classification," *ICCIDS 2019 - 2nd Int. Conf. Comput. Intell. Data Sci. Proc.*, pp. 1–6, 2019, doi: 10.1109/ICCIDS.2019.8862033.
- [13] R. A. Pramunendar, S. Wibirama, P. I. Santosa, P. N. Andono, and M. A. Soeleman, "A robust image enhancement techniques for underwater fish classification in marine environment," *Int. J. Intell. Eng. Syst.*, vol. 12, no. 5, pp. 116–129, 2019, doi: 10.22266/ijies2019.1031.12.
- [14] F. Optimization, "Forecasting Daily Stock Trends Using Random," pp. 1152–1155, 2019.
- [15] P. Agrawal, H. F. Abutarboush, T. Ganesh, and A. W. Mohamed, "Metaheuristic Algorithms on feature selection: A survey of one decade of research (2009-2019)," *IEEE Access*, vol. 9, pp. 1–1, 2021, doi: 10.1109/access.2021.3056407.